

Causal Machine Learning Project

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Exercise 1

The DAG in **Figure 1** illustrates the assumed causal relationship between 401(k) participation and net financial assets. The list of variables is as follows. Y : Net financial assets (the outcome); A : Eligibility for participating in a 401(k) plan (eligibility); P : Actual participation in the 401(k) plan; L : Income (an observed covariate); S : Propensity to save (unobserved).

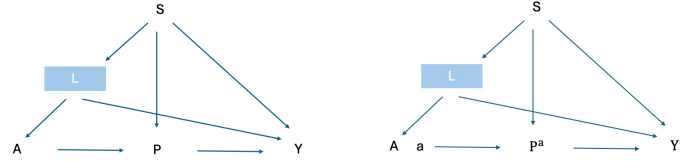


Figure 1: Causal framework illustrating: (a) the proposed Directed Acyclic Graph (DAG) depicting the causal relationships in our setting when conditioning on L ; and (b) the corresponding Single World Intervention Graph (SWIG) under intervention on variable A , showing that the conditional exchangeability assumption holds

This DAG allows for the most general case where the propensity to save, S , may affect eligibility (through income), participation and net assets. Note that S can be conceptualized to include every unobserved variable that may affect A through L . This setting implies that there would be confounding bias if we wanted to estimate the effect of P on Y . However, we can sort this by changing the focus to the effect of A on Y . If the reasoning that S affects Y only through the mediation of L holds, then there exists a backdoor path between A and Y through L that we can block by conditioning on L . This would result in S no longer being associated with A and therefore not acting as a confounder for the estimation of the effect of A on Y . Thus, conditional exchangeability, $Y^a \perp\!\!\!\perp A \mid L$, holds.

Exercise 2

We estimated nuisance parameters for the outcome and treatment models using SuperLearner with 10-fold cross-fitting. The library used in this and the following questions will be composed of the following learners: SL.speedglm, SL.earth, SL.ranger, and SL.gam. The plots display the distribution of estimated propensity scores and inverse probability weights (IPW) by treatment group. Boxplots are used to assess covariate overlap and identify potential weight instability, supporting diagnostics for the positivity assumption (see **Figure 2**).

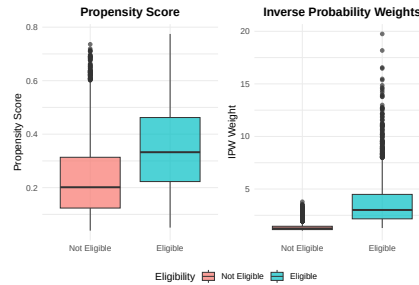


Figure 2: Distribution of propensity score and IPW Weight

The propensity score distributions exhibit some degree of overlap between the two groups. However, the distributions are clearly shifted, indicating systematic differences in covariates between eligible and non-eligible individuals.

Additionally, although the inverse probability weights (IPW) for the treated group exhibit some skewness, the maximum weight value is approximately 20 in a sample of 9915 observations, with only four other weights being higher than 15. Since there is no universally accepted threshold, it is hard to decide if the positivity assumption is violated. In any case, we consider that this a non-extreme setting and that it does not raise major concerns about the estimator instability, so ATE should be quite reliably estimated and there would be no need to restrict ourselves to the estimation of the ATT (as it is common practice when positivity

does not hold). However, using IPW or TMLE instead of IPW is still a good idea, so we benefit from the double-robustness property and we mitigate our dependency on the stability of the propensity score.

Table 1: Summary of ATE and ATT Estimates with 95% Confidence Intervals

Estimator	Estimate	SE	CI_Lower	CI_Upper
AIPW ATE	11459.28	1101.39	9300.57	13618.00
TMLE ATE	11455.85	1101.39	9297.17	13614.53
AIPW ATT	15371.33	1393.04	12640.97	18101.69
TMLE ATT	15964.87	1900.39	12240.18	19689.56
Naive ATE-ATT (without Covariate Adjustment)	27371.58	1681.56	24075.73	30667.43

Considering the estimates adjusted for covariates, the results are coherent across both AIPW and TMLE, with ATE estimates around 11,300 and ATT estimates around 15,300. The higher ATT values indicate that the treatment effect is stronger for individuals who were actually eligible, a result that seems plausible as people with better job benefits and conditions would also profit more from a tax-advantaged saving plan.

As predicted by theory, in the naïve implementation without covariate adjustment, the AIPW and TMLE estimators for both ATT and ATE, as well as the simple difference in means, all produce identical results. This happens because both the propensity score and outcome models are specified as constants, causing the estimators to reduce to the basic difference in group means.

Compared to the naïve ATE estimate of 27,372 ($SE = 1,682$), the AIPW and TMLE estimates are substantially lower, around 11,300 for ATE and 15,370 for ATT. This large difference highlights the importance of adjusting for confounding variables. The naïve estimator likely overstates the treatment effect because it does not account for baseline differences between eligible and non-eligible individuals.

Exercise 3

Because the efficient influence function of $\theta = E(Cov(A, Y|L))$ is given by

$$(A - E(A|L))(Y - E(Y|L)) - \theta$$

we can estimate θ by setting :

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n (A_i - E(A_i|W_i))(Y_i - E(Y_i|W_i))$$

Table 2: Estimation of $E[Cov(A, Y | L)]$ with 95% Confidence Interval

Parameter	Estimate	SE	CI_Lower	CI_Upper	p_value
$E[Cov(A, Y L)]$	8.5227	243.2896	-468.325	485.3704	0.9721

Based on the estimated conditional covariance $E[Cov(A, Y | L)]$, we find no statistical evidence of a residual association between 401(k) eligibility (A) and net financial assets (Y) after adjusting for covariates (X). The estimate is positive (8.52), with a large standard error (243.29) and a wide 95% confidence interval that includes zero (-468.33 to 485.37). The p-value (0.9721) strongly suggests that any observed association is not statistically significant.

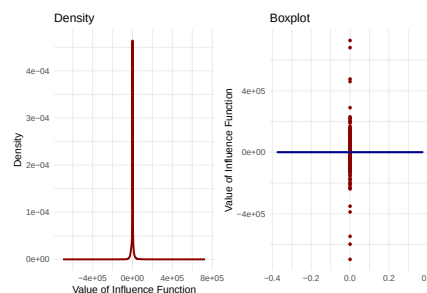


Figure 3: Distribution of Influence Function

The plots in **Figure 3** clearly reveal that the distribution of the influence function is extremely peaked around zero but exhibits very heavy tails. The boxplot shows a large number of extreme outliers on both sides, confirming the presence of high-influence observations. These extreme values explain the inflated standard error and overly wide confidence interval observed when using the full influence curve.

Exercise 4

In this exercise we will use 10-fold cross-fitting to construct the pseudo outcome and the weights of the R-learner, respectively $\tilde{Y}_i = \frac{Y_i - \hat{E}(Y_i | X_i)}{A_i - \hat{E}(A_i | X_i)}$ and $W_i = (A_i - \hat{E}(A_i | X_i))^2$. To do so, for every given fold, we have relied on two SuperLearners to fit the propensity and the outcome models on the 9-training folds and using the whole set of covariates listed in question 2. From now on we will refer to this full set of variables as X . Then, we estimate a new Superlearner to predict these weighted pseudo-outcomes but only as a function of age, income and education. We refer to this subset as W and to these predictions by $\hat{\tau}(W)$. The orthogonality property ensures that the error in the estimation of the nuisance parameters, $\hat{E}(A_i | X_i)$ and $\hat{E}(Y_i | X_i)$, has a mitigated effect in the quality of our estimates. As n increases, we would expect our estimates to perform as well as if the real outcome and propensity models were used.

In **Figure 4** we show the histogram of the marginal distribution of our CATE estimates $\hat{\tau}(W)$. We also show the scatterplots when plotted against age, income and years of education. We include in them the line resulting from fitting a Simple Linear Regression with intercept and we show the coefficients estimates to quantitatively analyze the association with every variable. To allow for a more general relation we also add a LOESS line. Due to the presence of extreme values in the original variable of Y , our estimates $\hat{\tau}(W)$ also show some extreme behavior for the 1% lowest and highest quantiles of their marginal distribution, which complicated the readability of the plots and had too strong an influence on the regressions. Therefore we have removed these observations before creating the plots.

From studying the histogram we see that the effects of eligibility on savings show a mainly positive unimodal distribution, with median a bit lower than 5k gains. The distribution is positively skewed with a long right-tail that is cut from the exclusion of the top 1%. With respect to the scatterplots of the effect of eligibility vs the individual variables of W we observe a positive association for both income and age. This is hardly surprising, as we would expect people with higher incomes to benefit the most from a 401(k) plan. Older people usually have higher incomes and a longer time period to benefit from the plan so the positive trend is also expectable. However, following a similar reasoning, we would also expect a positive effect of education, which is not clearly seen in the plot.

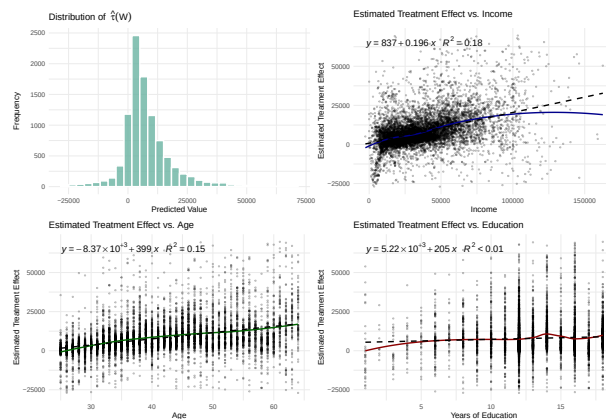


Figure 4: Marginal distribution of the R-learner CATE estimates and how they vary conditionally with income, age, and education

Exercise 5

We first compute the pseudo-outcome $C_i = \hat{Y}_i^1 + \frac{A_i}{\hat{p}_i} (Y_i - \hat{Y}_i^1)$ by performing a 10-fold cross-fitting where for each selected fold we fit the propensity and the outcome models on the training 9-folds to construct the C_i values on the selected one. For these models we employ the full set of covariates indicated in question 2. Then, we fit a model to predict the C_i using only age, income and years of education; and perform cross-fitting in a similar manner. We denote these predictions by $\tilde{Y}_1$. Our vector of estimates of $\tilde{Y}_1$ has mean around 22k dollars and a range that goes from -66k to 435k, with a highly positively skewed distribution

(see **Figure 5**). When we study the differences $Y - \tilde{Y}_1$ on the group of the treated, we see that it has a mean close to 0 ($\mu = 186$), but a very high volatility ($SD = 66136$).

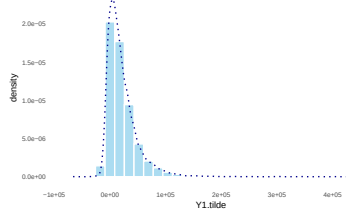


Figure 5: Histogram of $\tilde{Y}_1$

Now we are interested in estimating the MSE for $\tilde{Y}_1$, $\mathbb{E} \left[(Y_1 - \tilde{Y}_1)^2 \right]$, along with a 95% confidence interval.

We are targeting an expression where a counterfactual appears, so we need to find an estimand that we can identify. We can draw inspiration from G-computation and use that:

$$\mathbb{E} \left[(Y_1 - \tilde{Y}_1(W))^2 \right] = \mathbb{E}_X \left[\mathbb{E} \left[(Y - \tilde{Y}_1(W))^2 \mid X, A = 1 \right] \right],$$

where we rely on the conditional exchangeability and consistency assumptions (note that $\tilde{Y}_1$ is a deterministic function of W and therefore of X). This last expression is identifiable, we just need to fit a new model that predicts $(Y - \tilde{Y}_1)^2$ as a function of X on the subset of the treated. However this implies fitting another ML model to plug it in the average, which will carry plug-in bias. Another idea could be to mimic IPW derivation so as to fit:

$$\mathbb{E} \left[(Y_1 - \tilde{Y}_1(W))^2 \right] = \mathbb{E}_X \left[\frac{\mathbb{E}[A|X]}{\mathbb{E}[A|X]} \cdot \mathbb{E} \left[(Y_1 - \tilde{Y}_1(W))^2 \mid X \right] \right] = \mathbb{E} \left[\frac{A(Y - \tilde{Y}_1(W))^2}{P(A=1|X)} \right],$$

where we add positivity to the assumptions. We saw previously that while we did not have extreme cases, we did have several cases of weights over 10 and getting even to 20, which will affect the quality of the IPW estimate. This would suggest choosing a doubly-robust estimator like the AIPW or the TMLE. We opt for the first. Note that this will follow the same procedure of estimating the average of a counterfactual $E[Z_1]$ but where the outcome variable Z is the vector of squared differences $SD = (Y - \tilde{Y}_1)^2$.

Then our estimator becomes:

$$\hat{\psi}_{\text{AIPW}} = \frac{1}{n} \sum_{i=1}^n \left[\frac{A_i}{\hat{P}(A_i = 1 \mid X_i)} \left((Y_i - \tilde{Y}_1(W_i))^2 - \hat{\mathbb{E}} \left[(Y_i - \tilde{Y}_1(W_i))^2 \mid X_i, A_i = 1 \right] \right) + \hat{\mathbb{E}} \left[(Y_i - \tilde{Y}_1(W_i))^2 \mid X_i, A_i = 1 \right] \right].$$

We fit an AIPW estimator with 10-fold crossfitting as usual. The propensity score remains the same as in previous questions, but since SD is non-negative, we change the approach for the outcome model in that we take the logarithm as outcome for the initial modelling with the SuperLearner and then obtain our predictions by taking the exponential. Also, note that while we want to estimate the MSE of $\tilde{Y}_1$, which is a function of W , we should use all the control variables to fit both the propensity score model and the outcome regression model to meet the conditional ignorability condition.

Table 3: AIPW Estimate of MSE for $\tilde{Y}_1$ with 95% CI

Estimate	Std. Error	CI_Lower	CI_Upper
3101267498	634347930	1857945555	4344589441

The high value of the estimate is due to the very high volatility of both Y and the estimates $\tilde{Y}_1$. Even if we saw evidence pointing that our pseudo-outcomes are close to being unbiased, the existence of extreme values in Y_1 , added to the error dragged by the multiple models that we fitted and the low number of variables that we used to compute $\tilde{Y}_1$, explain the sometimes extreme values of the vector of differences (even when restricting to the treated group). By taking the square we exacerbate these extreme values, which leads to our high MSE estimate.

Regarding the division of work, Michelina Pizzocaro completed Exercises 2 and 3, while Carlos Rodriguez Ameal completed Exercises 4 and 5. Exercise 1 was done jointly by both students. Each report section was written by the student who completed the corresponding exercises.